

1 Introduction

We will discuss three main formulations for quantum mechanics:

- matrix mechanics (Dirac notation, commutation relations, basis independent),

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle; \quad \hat{H} |n\rangle = E_n |n\rangle, \quad (1)$$

for $|\psi(t)\rangle$ and $|n\rangle$ the time-dependent state and the time-independent, stationary eigenstates, respectively. This is the most abstract but also most general formulation because we need not commit to any particular basis. Some problems require us to eventually choose a specific basis to perform calculations, but not always.

- wave mechanics (Schrödinger wave equation, wave function, position basis):

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = \hat{H} \psi(\mathbf{r}, t), \quad \hat{H} \psi_n(\mathbf{r}) = E_n \psi_n(\mathbf{r}), \quad (2)$$

for $\psi(\mathbf{r}, t)$ and $\psi_n(\mathbf{r})$ the time-dependent wavefunction and the time-independent, stationary eigenfunctions, respectively. Inner products are integrals such as $\int \psi^*(\mathbf{r}, t) \phi(\mathbf{r}, t) d^3\mathbf{r}$ and expectation values are computed as $\int \psi^*(\mathbf{r}, t) \hat{O} \psi(\mathbf{r}, t) d^3\mathbf{r}$ for an operator $\hat{O}$. There is also an equivalent formulation in momentum space, where the wavefunction depends on momentum coordinates rather than position coordinates, all of this from Fourier transforming the position-space wavefunction.

- path integrals (Feynman formulation and the principle of least action),

$$\mathcal{K}(x_f, t_f; x_i, t_i) = \int \mathcal{D}[x(t)] e^{\frac{i}{\hbar} S[x(t)]}, \quad (3)$$

where $\mathcal{K}(x_f, t_f; x_i, t_i)$ is called the propagator or path integral, $x(t)$ the path, and $S[x(t)]$ the **action** functional. The integral is a functional integral over all possible paths $x(t)$. It is strongly oscillatory due to the factor $e^{\frac{i}{\hbar} S[x(t)]}$, such that contributions away from the extremal paths (those that make the action stationary, $\delta S[x(t)] = 0$, where the variation is with respect to the path $x(t)$) tend to cancel out. We will analyze this in a lot more detail later.

In this course we will discuss all three approaches, but for the most part we will be using matrix mechanics because it is the most economical and elegant formalism to use. You have probably already seen wave mechanics in your undergraduate course, it is what the first introductory chapters of Griffiths do. Path integrals are rarely covered in undergraduate courses, but they are a very powerful tool and extremely useful for more advanced courses in quantum mechanics, quantum field theory, and statistical mechanics (we will temporarily turn to Shankar's book when discussing path integrals).

Sakurai and Napolitano (S&N)'s book, like us, will use Dirac notation throughout. It is important you understand: what do bras and kets mean, in what space do they live in, what kind of operations can you perform on them, how to transpose and conjugate them, when you are allowed to “pass them through” other kets and bras and operators, how to project them onto each other, and so on. See previous notes.

1.1 Quantum Systems

QM describes very small systems, or very cold systems, or very isolated systems. It is a non-relativistic theory, so it is incomplete in the limit of high velocities/energies or strong gravitational fields. Neither of those are typical for atomic, molecular, or optical systems, which are the

primary focus of this course. Condensed matter is an interesting exception. Condensed matter systems are non-relativistic, but they involve collective excitations and a variable number of particles and quasi-particles. The formalism we develop here is called “first quantization” and it assumes the number of particles in a system is fixed, so it is no good to describe several many-body problems. Next semester we will see second quantization and Fock spaces, which despite still belonging to non-relativistic QM, gives us greater flexibility to deal with variable number of particles. The relativistic generalization of QM is called Quantum Field Theory (QFT) and we will see some of its origins next next semester.

Some more and other less obvious quantum systems include:

- molecules, atoms, and nuclei,
- electrons in solids (semiconductors, metals, superconductors),
- photons and other elementary particles,
- quantum dots, quantum wells, and other nanostructures,
- cold atoms and trapped ions in optical and electromagnetic traps,
- neutrinos traveling hundreds of kilometers through the Earth.

The entries in this list have very little in common as far as their energy scales, sizes, or the forces that hold them together. Each field has its own mathematical “technology” and jargon, but at the end of the day, they all rely on the same concepts of a Hilbert space and operators.



Reading: S&N Sections 1.1 and 1.2 for what follows. If you have never seen Dirac notation before, or if you are not comfortable with it, read Shankar Chapter 1 (the linear algebra) before you read Sakurai.

1.2 The Two-State System

Before we talk about where quantum mechanics is used, let me start with the single most useful and ubiquitous example: the two-state system. It is also what Sakurai calls “least classical, most quantum-mechanical system”. A remarkable fraction of modern quantum physics and most of the examples we encounter below live in a Hilbert space of only two dimensions, $\mathcal{H} = \mathbb{C}^2$.

Pick an orthonormal basis $\{|+\rangle, |-\rangle\}$:

$$\langle +|+\rangle = \langle -|-\rangle = 1, \quad \langle +|-\rangle = 0, \quad |+\rangle \langle +| + |-\rangle \langle -| = \mathbb{1}. \quad (4)$$

The last equation is just the completeness relation for this 2-d Hilbert space. The most general (pure) state is a normalized superposition,

$$|\psi\rangle = a|+\rangle + b|-\rangle, \quad |a|^2 + |b|^2 = 1, \quad (5)$$

with $a = \langle +|\psi\rangle$ and $b = \langle -|\psi\rangle$. An overall phase is not measurable, so we can always choose a to be real and non-negative and write

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|+\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|-\rangle, \quad (6)$$

so that the physically distinct states are labelled by a point (θ, φ) on a sphere, S^2 . This is called the **Bloch sphere**, and you will meet it again and again.



Note the $\theta/2$. Show that states that states sitting at opposite points on the sphere, θ and $\pi - \theta$ with $\varphi \rightarrow \varphi + \pi$ are orthogonal in $\mathcal{H}$ (i.e., $\langle \psi(\theta, \varphi) | \psi(\pi - \theta, \varphi + \pi) \rangle = 0$).

Let us now look at what operators on $\mathbb{C}^2$ look like. Start by introducing the Pauli operators:

$$\hat{\sigma}_x = |+\rangle\langle-| + |- \rangle\langle+|, \quad \hat{\sigma}_y = -i|+\rangle\langle-| + i|- \rangle\langle+|, \quad \hat{\sigma}_z = |+\rangle\langle+| - |- \rangle\langle-|, \quad (7)$$

If we represent $|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, then the matrix elements of the Pauli operators in this basis can be calculated as

$$(\sigma_x)_{mn} = \langle m | \hat{\sigma}_x | n \rangle, \quad m, n \in \{+, -\}, \quad (8)$$

giving the familiar matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (9)$$

It turns out that any operator on a two-state system can be represented by a 2×2 matrix, and any of them can be written in terms of the identity and the three Pauli operators,¹

$$\hat{O} = c_0 \hat{1} + c_x \hat{\sigma}_x + c_y \hat{\sigma}_y + c_z \hat{\sigma}_z, \quad c_0, c_x, c_y, c_z \in \mathbb{C}. \quad (10)$$

You should be interpreting Eq. (7) and Eq. (9) as the same object: one in what is called the “dyadic” form and the other in matrix form. The dyadic form tells you what the operator does (“ σ_x turns a plus into a minus and vice-versa”) and the matrix form is what it looks like in a specific basis. They obey

$$\sigma_i \sigma_j = \delta_{ij} \hat{1} + i \epsilon_{ijk} \sigma_k, \quad (11)$$

with ϵ_{ijk} the Levi-Civita symbol ($\epsilon_{123} = 1$ and completely antisymmetric), as well as

$$[\sigma_i, \sigma_j] = \sigma_i \sigma_j - \sigma_j \sigma_i = 2i \epsilon_{ijk} \sigma_k. \quad (12)$$

The commutation relation between operators is said to define the “algebra” of the operators. In this case, the (Lie) algebra is called the **Pauli algebra** or the **su(2) algebra**.² This is the fundamental algebra of angular momentum in quantum mechanics. We will see later one that this is because the su(2) algebra is isomorphic to so(3), the algebra of rotations in three-dimensional space (btw, this is the reason behind the factor of 1/2 in the argument of the signs in Eq. (6)). Angular momentum operators are intimately connected to rotations in three dimensions (we will say they are the generators of rotations).

Let us imagine our two state system is just the spin states of a spin-1/2 particle. We then define the **spin operators** as

$$\hat{S}_i = \frac{\hbar}{2} \hat{\sigma}_i \quad (13)$$

and the algebra becomes $[\hat{S}_i, \hat{S}_j] = i\hbar \epsilon_{ijk} \hat{S}_k$. Spin-1/2 systems are described by the Pauli operators. We may have particles with larger spin. In that case the spin operators will still obey the su(2) algebra, but they will have a different matrix representation.

¹This is just the fact that any 2×2 matrix can be expressed as a linear combination of $\hat{1}$, σ_x , σ_y , and σ_z .

²The Lie *algebra* su(2) is the algebra of the generators of the Lie *group* SU(2), the special unitary group of degree 2, which consists of all 2×2 (degree 2) unitary matrices with determinant 1 (special).



For two-state systems, we can make very general statements about the Hamiltonian. Since $\mathbb{1}$ contributes only an overall phase to the time evolution (as we will see, this is physically irrelevant), the most general two-state Hamiltonian is

$$\hat{H} = \frac{\hbar\Omega}{2}(\vec{n} \cdot \vec{\sigma}) = \frac{\hbar\Omega}{2}(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z), \quad (14)$$

with $\vec{n} = (n_x, n_y, n_z)$ a real unit vector, $\vec{\sigma} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$, and Ω a frequency. You can show that the time evolution generated by this Hamiltonian is a rotation of the Bloch vector about the axis $\vec{n}$ at an angular frequency Ω . This is the general statement behind spin precession, Rabi flopping, neutrino oscillations, atomic clock, etc. **Btw, the $\vec{v}$ versus bold vectors $\mathbf{v}$ notation is interchangeable throughout these notes. I cannot use bold greek letters (σ) with accessibility packages, and trying to use $\vec{v}$ everywhere would be super cumbersome with $\hat{v}$, so here we are.**

1.2.1 Our First Hamiltonian

We know that the system's Hamiltonian is an energy operator. It determines the energy of the system's states and, through the Schrödinger equation, governs their time evolution. In two-state systems, we only have the Pauli operators, so it makes sense to use those to build a Hamiltonian. All we need is a dimensionful parameter and the specific direction $\mathbf{n}$ we choose (or equivalently: what specific combination of Pauli matrices we believe our system is described by).

Let us start with the simplest choice: aligning the Hamiltonian along the z -axis of the Bloch sphere:

$$\hat{H} = \frac{\hbar\omega_0}{2}\sigma_z, \quad (15)$$

I am adding the $\hbar$ explicitly, but in natural units $\hbar = 1$ and therefore ω_0 must have units of energy. The energy eigenstates and eigenvalues of this Hamiltonian are

$$\hat{H}|\pm\rangle = \pm\frac{\hbar\omega_0}{2}|\pm\rangle. \quad (16)$$

The Hamiltonian in (15) describes, among many other two-state systems, a spin- $\frac{1}{2}$ particle in a magnetic field along the z -axis. In that case, $\omega_0 = -\gamma B$ is related to the strength of the magnetic field B and the particle's gyromagnetic ratio γ , and the two energy eigenstates $|\pm\rangle$ would correspond to spin pointing up or down along the z -axis.

As we will prove soon, the time evolution of a quantum state is governed by the Schrödinger equation,

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle. \quad (17)$$

For a time-independent Hamiltonian like (15), the solutions to this equation is very simple:

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\psi(0)\rangle. \quad (18)$$

The best way to interpret that exponential of an operator is to Taylor expand it as a power series:

$$e^{-\frac{i}{\hbar}\hat{H}t} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar}\hat{H}t\right)^n = \hat{I} - \frac{i}{\hbar}\hat{H}t + \frac{1}{2!} \left(-\frac{i}{\hbar}\hat{H}t\right)^2 + \dots \quad (19)$$

where now each term is a sequence of operators acting on the state $|\psi(0)\rangle$. For our two-state system, the time evolution of the basis kets is straightforward, we just hit the basis kets with the Hamiltonian to return the eigenvalues ($\pm\frac{\hbar\omega_0}{2}$) and exponentiate again to get:

$$|+(t)\rangle = e^{-i\omega_0 t/2}|+\rangle, \quad |-(t)\rangle = e^{+i\omega_0 t/2}|-\rangle. \quad (20)$$

Now, start with the state

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle). \quad (21)$$

Time evolution the basis kets then gives

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}(e^{-i\omega_0 t/2}|+\rangle + e^{+i\omega_0 t/2}|-\rangle) = \frac{e^{-i\omega_0 t/2}}{\sqrt{2}}(|+\rangle + e^{i\omega_0 t}|-\rangle). \quad (22)$$

That is, the superposition of the basis kets acquires a relative phase that evolves in time. Recall that the overall global phase $e^{-i\omega_0 t/2}$ does not affect any physical observables.

Now we want to measure this state. If it was a spin-1/2 system, we can say we want to measure the direction of the spin. In this case, the relevant observables are the spin operators $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$ (all hermitian operators).

Having set this up, I want to show how the question you ask in quantum mechanics is extremely important. Devising the right measurement and the right observable is most of the work.

Say you only want to know what fraction of the time the system is found in the $|+\rangle$ state versus the $|-\rangle$ state (spin up versus spin down, for example). In that case, you'd be tempted to just calculate the probabilities of finding the system in the $|+\rangle$ and $|-\rangle$ states:

$$|\langle +|\psi(t)\rangle|^2 = \frac{1}{2}, \quad |\langle -|\psi(t)\rangle|^2 = \frac{1}{2}. \quad (23)$$

These are constant in time. We started with an even superposition of the $|+\rangle$ and $|-\rangle$ states and we remained with an even population of each over time. However, this is not the full story of the dynamics of the system.

Far more interesting is to calculate the expectation values of the spin operators $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$. They tell you the average outcome of measuring the corresponding component of the spin. We do this by calculating the expectation values of the spin operators on the state $|\psi(t)\rangle$:

$$\langle \hat{S}_i \rangle_{\psi(t)} \equiv \langle \psi(t) | \hat{S}_i | \psi(t) \rangle, \quad i = x, y, z. \quad (24)$$



Note I am putting the time dependence explicitly in ψ . As we will see later, this is equivalent to transferring it to the operators themselves: $\langle \hat{S}_i \rangle_{\psi(t)} = \langle \hat{S}_i(t) \rangle_{\psi}$.

Carrying this out (either using the dyad form or the matrix representation with Pauli matrices) gives

$$\langle \hat{S}_x \rangle_{\psi(t)} = \frac{\hbar}{2} \cos(\omega_0 t), \quad \langle \hat{S}_y \rangle_{\psi(t)} = \frac{\hbar}{2} \sin(\omega_0 t), \quad \langle \hat{S}_z \rangle_{\psi(t)} = 0. \quad (25)$$

The expectation values of the spin components is changing with time, even though the populations in the $|+\rangle$ and $|-\rangle$ states remain constant. Representing the expectation values as a vector, we have the Bloch vectors

$$\mathbf{S}(t=0) = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{S}(t) = \begin{pmatrix} \langle \hat{S}_x \rangle_{\psi(t)} \\ \langle \hat{S}_y \rangle_{\psi(t)} \\ \langle \hat{S}_z \rangle_{\psi(t)} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \cos(\omega_0 t) \\ \sin(\omega_0 t) \\ 0 \end{pmatrix}. \quad (26)$$

This vector is rotating in the xy -plane with angular frequency ω_0 , tracing out a circle of radius $\hbar/2$. It starts out pointing along the x -axis at $t = 0$, and can end up on the diametrically opposite side of the circle at $t = \pi/\omega_0$. In other words, the Bloch vector precesses in the equatorial plane at frequency ω_0 .

This is the physics of a spin-1/2 particle with a magnetic moment in a magnetic field, $\hat{H} = -\boldsymbol{\mu} \cdot \mathbf{B}$ with $\boldsymbol{\mu} = \gamma \hat{\mathbf{S}}$ and $\mathbf{B} = B\hat{z}$, this is Larmor precession at $\omega_0 = -\gamma B$, and it is the physics behind Nuclear Magnetic Resonance (NMR) and Magnetic Resonance Imaging (MRI). MRIs work by detecting the precession of nuclear spins in a strong magnetic field, using radiofrequency pulses to manipulate the spins and measure their response. For a proton, $\gamma/2\pi \simeq 42.6 \text{ MHz/T}$. A typical 1.5 T magnet precesses the hydrogen nuclei in your body emitting radio waves (64 MHz is about 5 meter wavelength radio waves).



Sakurai goes into spin precession in Section 2.1.5, and again from the point of view of rotations in Section 3.2.2. Next semester we will come back to it again to study the effect of an external magnetic field driver, with a time-dependent Hamiltonian.



Verify Eq. (25) directly from Eq. (22) using the dyadic forms in Eq. (7). Then show that for a general initial state written as in Eq. (6), the effect of the evolution is simply $\varphi \rightarrow \varphi + \omega_0 t$ at fixed θ .

Added after class:

2 Uncertainty on $\hat{S}_x$ and $\hat{S}_y$

Let us address the incompatibility of $\hat{S}_x$, $\hat{S}_y$ and $\hat{S}_z$ in more detail. Take a spin prepared exactly parallel to $\hat{z}$, that is, the state $|+\rangle$. The uncertainty of an observable in a state $|\psi\rangle$ is

$$(\Delta O)^2 = \langle \hat{O}^2 \rangle_\psi - \langle \hat{O} \rangle_\psi^2. \quad (27)$$

The right nomenclature of the uncertainty is: $(\Delta O)^2$ is called the **variance** of the observable $\hat{O}$, ΔO is called the **standard deviation**, $\langle \hat{O}^2 \rangle_\psi$ is the **second moment** of the observable in the state $|\psi\rangle$, and $\langle \hat{O} \rangle_\psi$ is the **first moment**. Since $\hat{S}_z |+\rangle = \frac{\hbar}{2} |+\rangle$,

$$\langle \hat{S}_z \rangle = \frac{\hbar}{2}, \quad \langle \hat{S}_z^2 \rangle = \frac{\hbar^2}{4} \quad \implies \quad (\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{4} = 0. \quad (28)$$

The expectation value of the operator on its eigenstate always vanishes. If we know we have prepared the spin exactly along $\hat{z}$, then the uncertainty in S_z is zero. That is not the end of the story though.

Now do the same for $\hat{S}_x$. One can show that

$$\hat{S}_x |\pm\rangle = \frac{\hbar}{2} |\mp\rangle \quad \implies \quad \langle \hat{S}_x \rangle = \langle + | \hat{S}_x | + \rangle = \frac{\hbar}{2} \langle + | - \rangle = 0. \quad (29)$$

The second moment is much simpler:

$$\hat{S}_x^2 = \frac{\hbar^2}{4} \hat{\sigma}_x^2 = \frac{\hbar^2}{4} \mathbb{1}. \quad (30)$$

Note that any $\hat{S}_{x,y,z}^2$ are proportional to the identity, so $\langle \hat{S}_{x,y,z}^2 \rangle = \hbar^2/4$ for every state (it never vanishes). Putting it all together, one can show that

$$\Delta S_x = \Delta S_y = \frac{\hbar}{2} \quad (31)$$

Now let's show that Eq. (31) is not just nonzero, it is the actually the largest value possible. Note that measuring S_x on the state $|+\rangle$ or $|-\rangle$, returns $+\hbar/2$ half the time and $-\hbar/2$ half the time. For any observable taking the values $\pm\hbar/2$ with probabilities p and $1-p$, we can show that $(\Delta O)^2 = \hbar^2 p(1-p)$, which is maximized at $p = \frac{1}{2}$. So, making S_z perfectly aligned has made S_x and S_y as random as they can possibly be.

There is an interesting generic observation about spin- $\frac{1}{2}$ systems. Later we will show that $\hat{\mathbf{S}}^2 = s(s+1)\hbar^2\mathbb{1} = \frac{3\hbar^2}{4}\mathbb{1}$, with $s = \frac{1}{2}$, so in **any** state

$$\langle \hat{\mathbf{S}}^2 \rangle = \langle \hat{S}_x^2 \rangle + \langle \hat{S}_y^2 \rangle + \langle \hat{S}_z^2 \rangle = \frac{3\hbar^2}{4}, \quad (32)$$

which also follows directly from Eq. (30) applied to each component. In addition, for any pure state the Bloch vector $\mathbf{S}$ has length $\hbar/2$, i.e.

$$|\mathbf{S}|^2 = \sum_i \langle \hat{S}_i \rangle^2 = \frac{\hbar^2}{4}. \quad (33)$$



Prove this by calculating it for the most general state of Eq. (6).

Subtracting Eq. (33) from Eq. (32), we find the following relation for pure states:

$$\sum_i (\Delta S_i)^2 = \frac{3\hbar^2}{4} - \frac{\hbar^2}{4} = \frac{\hbar^2}{2}. \quad (34)$$

This is called a sum rule, and it proves that the total variance is a constant of the two-state system. You are free to move it around between the three components, but you can't reduce it. You can never set all three variances to zero.

In our example in class, driving $\Delta S_z \rightarrow 0$ forced the whole $\hbar^2/2$ into the transverse directions with half into each, which is exactly Eq. (31).



Is there a state in which all three uncertainties, ΔS_x , ΔS_y , and ΔS_z , are equal? If so, find it and give its Bloch vector.